

VINAY REDDY PEDDIREDDY

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EDUCATION

Queen Mary University of London
MSc in Computer Game
Specialized skills

Greater London, UK
Sep 2023- Current

- ✦ Advanced Game Development, Fundamentals of Game Design, Computational Game Design
Interactive Agents and Procedural Generation, Computational Creativity, Level Design

Sathyabama Institute of Science and Technology
Bachelor's in Computer science

Chennai, India
Aug 2018- Sep 2022

- ✦ Top 1% of 2022 computer graduate students
- ✦ **21st** Rank in All India Wide Entrance Exam

RELEVANT EXPERIENCE

Indie Game Developer — Voxel Web Game

May 2025- Recent

London, UK

Stack parity: TypeScript-first, Babylon.js/WebGL, React, Vite, Node (basics), CSS3D, Blender, Blockbench, Figma, Git

- **Editable worlds, quickly:** Built a chunked voxel engine with **raycast**, **face-aware block edits** and **dirty-chunk rebuilds** for instant feedback—designed APIs so new block types and interactions can be added in a few lines.
- **Engine + gameplay systems:** Implemented **AABB physics** (grounded, slope, step), **FP/TP controller** with pointer-lock and stable yaw/pitch, and an **action system** decoupling input from gameplay logic for rapid iteration.
- **Reusable features for others:** Created an **FPS view-model layer** (hands/weapons on separate render pass) with a small **state machine**—drop-in ready for additional tools/weapons without clipping.
- **AI & agents:** Prototyped **NPC FSM** with block-aware stepping and animation sync—clean interfaces so behaviours can be extended by other devs.
- **Performance + profiling mindset:** **Greedy meshing**, batched geometry, **low-GC frame loop** (object reuse, cached lookups), and capped ray casts per frame; **toggleable diagnostics** to profile and tune in-engine.
- **Product sense & UX:** Designed mini-game flows (e.g., **King of the Hill**), quick UI in **CSS3D** panels/HUD, and ran fast playtests to balance roles/objectives; shipped art/UX polish (textures, flipbook break VFX).
- **Wore many hats:** Engine + gameplay + UI/UX + content (modelled/rigged assets in Blender/Block bench), docs and small READMEs for setup and contribution.

Game Designer (Intern)

FEB 2022- MAR 2023

Creative Monkey Games, Coimbatore, India

- Designed game mechanics, monetization models and interactive features in Unity, improving game functionality by **2x** and enhancing player engagement. Collaboration with cross-functional teams of developers and artists.
- Experienced in Live Ops. Debugged and resolved **50+** technical issues related to physics, gameplay, and interactivity, ensuring a stable and reliable gameplay experience for **500+** beta testers.
- Optimized game performance for iOS and Android devices, reducing load times and ensuring smooth gameplay across multiple devices.
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Game Design and Development Society Leader

JAN 2019- Dec 2021

Sathyabama Institute of Science and Technology, Chennai

- Founded and led a game design and development community, fostering creativity and collaboration among students.
- Organized 8+ events per semester, including workshops ("Build Your Own Mini-Game"), lectures ("The Psychology Behind Games"), and team-based game design summits.
- Collaborated with local schools to introduce gamified learning initiatives, promoting student engagement through interactive educational techniques.
- **Key Achievement:** Mentored 100+ students in game development, improving their technical and collaborative skills.

ACADEMIC PROJECTS

Queen Mary University of London

2023-2024

Designing Mechanics and Optimizing AI Agents in Tabletop Games Framework (TAG) ○ Designed and implemented new game mechanics for competitive board games within the Tabletop Games Framework (TAG), focusing on enhancing player fairness, mastery, and gameplay balance in games like Connect 4 and Poker Texas Hold'em.

- Developed and integrated new AI agents, leveraging TAG's API and tools to optimize agent performance through parameter tuning, improving decision-making and strategies in competitive environments.
- Conducted live operations tasks, including bug fixing, adding new mechanics, and A/B testing, enhancing overall game functionality and player experience.
- Achieved a **1.6x** increase in win rate by enhancing MCTS-based AI agents and received positive feedback from **31** play testers on the new mechanics, confirming improvements in game balance and player engagement.

Advanced Game Development Using C++ ○ Design playable game using advanced game programming patterns and subsystems that optimize performance, memory usage, and system communication in a game engine with Real-Time Log Analysis Tool.

- **41%** memory consumption reduced and **3x** faster performance were measurable by the efficient execution of range of programming principles and patterns using C++
- These results were achieved by incorporating a Big Array ECS architecture, integrating Flyweight, Observer, and Service Locator design patterns, and implementing the Command pattern for modular and flexible system execution.

Implementing Machine Learning for Game Development ○ Analysed player death patterns in Battleground Game, using data-driven approaches to optimize game balance and guide future game updates, aligning with computational and data-oriented game design practices.

- Delivered player behaviour insights leading to **15%** improvement in game updates and map design efficiency.
- Conducted feature engineering and clustering analysis on DotA players, providing insights into player behaviours and improving data understanding.
- Identified more than **5** unique player profiles, enhancing in-game analytics and personalizing player engagement strategies.

Procedural Animation for Sword-Fighting Simulator

- Designed a dynamic animation system for sword-fighting combat, incorporating second-order dynamics in Unity 3D to enhance visual realism and create a natural sense of inertia in sword movements.
- Integrated advanced mathematical concepts, inverse kinematics, an innovative input system for responsive movement, and a particle system to improve combat interactions.
- Received **100%** positive feedback from **21** play testers for the fluidity and realism of dynamic sword movements, elevating the overall combat experience.

Interactive Agents and Procedural Generation ○ Developed a dynamic 2D Zombie Apocalypse simulation in Unity with AI agents driven by behaviour trees. ○ Designed a procedural generation pipeline using cellular automata, increasing content diversity by **20x** and reducing decision-making time of Interactive agents by **15-25%**.

SKILLS

Programming Languages: TypeScript/JavaScript, C++, C#, Java, and python

Tools & Frameworks: WebGL, BabylonJS, three.js, React, BlockBench, Blender, Unity, Unreal, Git, Node

Game Mechanics & System Design: Web Game development, Procedural generation, data-driven game design, AI systems, and dynamic simulations.

Problem-Solving: Strong analytical skills with experience in designing complex mechanics for Game Designing.

Team Collaboration: Experience working using GitHub and team-based environments.

Game Designing: Passionate about Level Design, dynamic storytelling, and large-scale simulations.